Dispute copilot on AWS

Target deployment for a bank-hosted run · serverless, human-gated, single region

TARGET STATE · NOT BUILT

USER
Customer / contact centre
web · email · phone
untrusted input
no direct model access

BANK
Bank evidence APIs
transactions · merchant
read-only snapshots
source_id preserved

BEDROCK ONLY · NO EXTERNAL PROVIDER
Approved model ID only.
Model off by default. No vector DB.

HUMAN
Reviewer
fraud ops · MFA
approve or reject
cannot be the agent

BANK
Bank case system
POST /records
409 duplicate
system of record

01 INTAKE
validate · buffer · store

APIGW
Private intake API
API Gateway · REST
POST /v1/cases
IAM auth · 413 limit

SQS
Intake queue
SQS · DLQ
buffer · 3 receives
then dead-letter

λ
Validate intake
build_records.py
schema · size · source IDs
reject 400/413

S3
Raw intake + files
S3 · SSE-KMS
versioned · Block Public
no public access

Only the reviewer console is public.
The agent has no inbound internet path.

02 ORCHESTRATE & BUILD
workflow · tools · policy

SFN
DisputeCaseWorkflow
Step Functions · Standard
retry · timeout · task token
orchestrates every stage

λ
Build working record
contract_records.py
policy by reported date
transactions + merchant

INT
Bank tool gateway
tools/runtime.py · client
3 attempts · 30s timeout
circuit breaker · audit

S3
Policy + procedure
S3 · versioned corpus
Object Lock · exact quotes
V1 / V2 by reported date

PrivateLink / HTTPS to the bank.
No NAT for bank traffic.

03 DECIDE
rules first · model at the gap

λ
Rules draft
contract.py
classification · routing
deadline · citations

λ
Model gateway
model_cue.py · model_clients.py
two closed questions
schema-validated answer

BR
Bedrock + Guardrails
Amazon Bedrock
approved model only
single turn · no tools

DDB
Model registry
DynamoDB · models.toml
schema hash · approved_by
live call refused if empty

The model only names a type or answers
a yes/no. Code sets every field.

task calls

04 REVIEW
human gate · hash-bound

CF
Reviewer console
CloudFront · WAF · Cognito
MFA · reviewers group
only public surface

APIGW
Review API
API Gateway · Cognito
approve · reject
draft hash

λ
Review service
tools/runtime.py · review
conditional write
SendTaskSuccess

DDB
Reviews table
DynamoDB
status · reviewer
draft_hash · PITR

The workflow role can read a review
but cannot write an approval.

05 CREATE
idempotent · create-only

λ
Create approved case
tools/runtime.py · records
Idempotency-Key
conditional create

DDB
Idempotency + cases
DynamoDB
key → response · TTL
duplicate → 409

S3
Draft + contract
S3 · Object Lock
KMS · no prompts or PANs
audit export

λ
Audit emitter
EventBridge · DynamoDB Streams
case events · S3 Object Lock
no prompts or PANs

Create only. No update, close or delete.
A changed draft needs a new approval.

PLATFORM · CROSS-CUTTING

IAM
Identity & keys
KMS · IAM · Secrets Manager
per-function roles
VPC endpoints · no NAT
no secrets in env

CT
Audit & detection
CloudTrail · Config
GuardDuty · Macie
S3 Object Lock
data events on S3/DDB

CW
Observability
CloudWatch · X-Ray
structured logs · metrics
alarms · dashboards
no prompts or PANs

GH
Delivery
GitHub Actions OIDC · CDK
pytest 100% · mypy · ruff
dev / staging / prod
no long-lived keys

AWS CLOUD · US-EAST-1 · SINGLE REGION

no public agent endpoint

Target state, not the assessed deployment.
The repo today runs one local container with SQLite and no reviewer identity. This is the proposed AWS shape for a bank-hosted run; it is not built or measured.

synchronous call · event / async · λ AWS service

All compute in private subnets. CloudFront + WAF is the only public surface.

No model autonomy: no tools, no browsing, no memory.